Trials@uspto.gov
571-272-7822

Paper 41
Date: April 19, 2023

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

JSR CORPORATION and JSR LIFE SCIENCES, LLC,
Petitioner,

v.

CYTIVA BIOPROCESS R&D AB,
Patent Owner.

IPR2022-00036
IPR2022-00043
Patent 10,213,765 B2

Before ULRIKE W. JENKS, SHERIDAN K. SNEDDEN, and SUSAN L. C. MITCHELL, *Administrative Patent Judges.*

JENKS, *Administrative Patent Judge*.

JUDGMENT
Consolidated Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review of claims 1–7, 10–20, and 23–26 ("the challenged claims") of U.S. Patent No. 10,213,765 B2 (Ex. 1001, "the '765 patent"). We have jurisdiction under 35 U.S.C. § 6, and enter this Consolidated Final Written Decision pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons set forth below, we determine that JSR Corporation and JSR Life Sciences, LLC (collectively, "Petitioner") has shown, by a preponderance of the evidence, that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e).

A. *Consolidated Proceedings*

The two captioned proceedings (IPR2022-00036 and IPR2022-00043 (or "the '043 IPR") involve the '765 patent and challenge the same set of claims. The asserted grounds and prior art contentions are different in each proceeding. Consolidation is appropriate where, as here, the Board can more efficiently handle the common issues and evidence, and also remain consistent across proceedings. Under 35 U.S.C. § 315(d), the Director may determine the manner in which these pending proceedings may proceed, including "providing for stay, transfer, consolidation, or termination of any such matter or proceeding." *See also* 37 C.F.R. § 42.4(a) ("The Board institutes the trial on behalf of the Director."). There is no specific Board Rule that governs consolidation of cases. But 37 C.F.R. § 42.5(a) allows the Board to determine a proper course of conduct in a proceeding for any situation not specifically covered by the rules and to enter non-final orders to administer the proceeding. Therefore, on behalf of the Director under § 315(d), and for a more efficient administration of these proceedings, we consolidate IPR2022-00036 and IPR2022-00043 for purposes of rendering this Final Written Decision.

B. *Procedural History*

JSR Corporation and JSR Life Sciences, LLC (collectively, "Petitioner") filed a Petition for an *inter partes* review of claims the challenged claims under 35 U.S.C. § 311 in each proceeding. Paper 1[1] ("Pet."). Petitioner supported the Petition with the Declaration of Dr. Steven M. Cramer. Ex. 1002. Cytiva Bioprocess R&D AB ("Patent Owner") filed a Patent Owner Preliminary Response to the Petition. Paper 9 ("Prelim. Resp.").

On April 21, 2022, pursuant to 35 U.S.C. § 314(a), we instituted trial ("Decision" or "Dec." (Paper 10)) to determine whether any challenged claim of the '765 patent is unpatentable.

In IPR2022-00036, Petitioner asserts the following grounds of unpatentability (Pet. 4):

Claim(s) Challenged	35 U.S.C. §[2]	Reference(s)/Basis
1–4, 12, 14–17, 25	103(a)	Linhult, Abrahmsén
1–7, 10–20, 23–26	103(a)	Linhult, Hober

[1] We note that the evidence filed in both proceedings is generally consistent in having the same exhibit number. Therefore, we reference exhibits and paper numbers as they appear in the record of IPR2022-00036, unless otherwise noted.

[2] The Leahy-Smith America Invents Act ("AIA") included revisions to 35 U.S.C. § 103 that became effective on March 16, 2013. Because the '765 patent issued from an application claims priority from an application filed before March 16, 2013, we apply the pre-AIA versions of the statutory bases for unpatentability.

Claim(s) Challenged	35 U.S.C. §[2]	Reference(s)/Basis
1–7, 10–20, 23–26	103(a)	Linhult, Abrahmsén, Hober
1–7, 10–20, 23–26	103(a)	Abrahmsén, Hober

In IPR2022-00043, Petitioner asserts the following grounds of unpatentability ('043 IPR Pet. 4):

Claim(s) Challenged	35 U.S.C. §[3]	Reference(s)/Basis
1-7, 10–20, 23–26	103(a)	Berg, Linhult
2, 3, 15, 16	103(a)	Berg, Linhult, Hober
1, 2, 5–7, 10–15, 18–20, 23–26	103(a)	Berg, Abrahmsén
2–4, 15–17	103(a)	Berg, Abrahmsén, Hober

Patent Owner filed a Patent Owner Response to the Petition. Paper 16 ("PO Resp."). Patent Owner supported the Response with the Declaration of Dr. Daniel Bracewell (Ex. 2025). *See* PO Resp., iv (Exhibit List). Petitioner filed a Reply to the Patent Owner Response. Paper 29 ("Reply"). Petitioner supported the Reply with a Reply Declaration from Dr. Steven M. Cramer. Ex. 1061. Patent Owner filed a Sur-reply to Petitioner's Reply. Paper 35 ("Sur-reply").

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[3] The Leahy-Smith America Invents Act ("AIA") included revisions to 35 U.S.C. § 103 that became effective on March 16, 2013. Because the '765 patent issued from an application claims priority from an application filed before March 16, 2013, we apply the pre-AIA versions of the statutory bases for unpatentability.

On February 16, 2023, the parties presented arguments at an oral hearing. Paper 36. The hearing transcript has been entered in the record. Paper 40 ("Tr.").

For the reasons set forth below, we determine that Petitioner has shown by a preponderance of the evidence that claims 1–7, 10–20, and 23–26 of the '765 patent are unpatentable.

C. *Real Parties in Interest*

Petitioner identifies itself, JSR Corporation and JSR Life Sciences, LLC, along with JSR Micro NV, as the real parties-in-interest. Pet. 2. Patent Owner identifies itself, Cytiva Bioprocess R&D AB, along with Cytiva Sweden AB and Danaher Corporation as real parties-in-interest. Paper 7, 1.

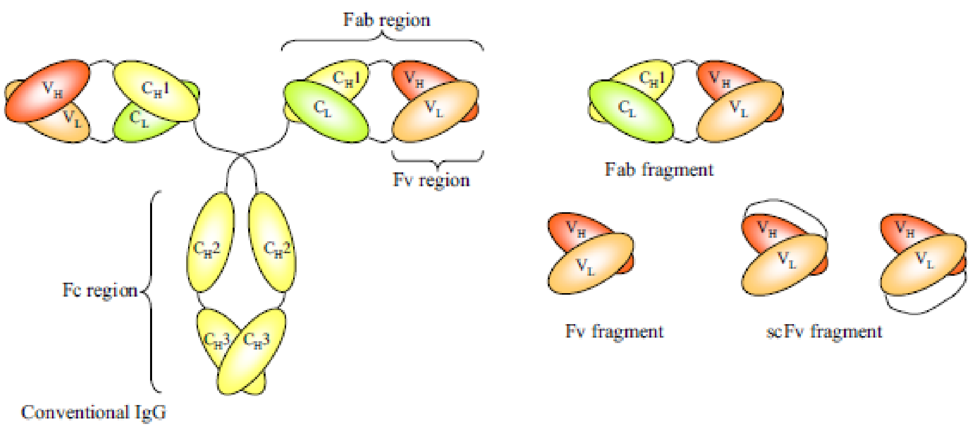
D. *Related Matters*

The '765 patent is at issue in *Cytiva BioProcess R&D et al. v. JSR Corp. et al.*, Civil Action No. 1:21-cv-00310 (D. Del.). Pet. 2; Paper 5, 1.

In addition to the '765 patent challenged here, Petitioner has filed Petitions for *inter partes* review of related U.S. patents as follows: U.S. Patent No. 10,343,142 B2 ("the '142 patent") in IPR2022-00041 and IPR2022-00044; and U.S. Patent No. 10,875,007 B2 ("the '007 patent") in IPR2022-00042 and IPR2022 00045. Pet. 2–3; Paper 7, 1–2. Petitioner indicates that the '142 patent and the '007 patent are also being asserted in the above-cited district court case. Pet. 3. The parties further list a pending application in the same family, U.S. App. Serial No. 17/107,600. Pet. 2; Paper 7, 2.

E. *Subject matter background*

Antibodies (also called immunoglobulins) are glycoproteins, which specifically recognize foreign molecules. These recognized foreign molecules are called antigens. Ex. 2001, 1. A schematic representation of the structure of a conventional IgG and fragments is shown below:

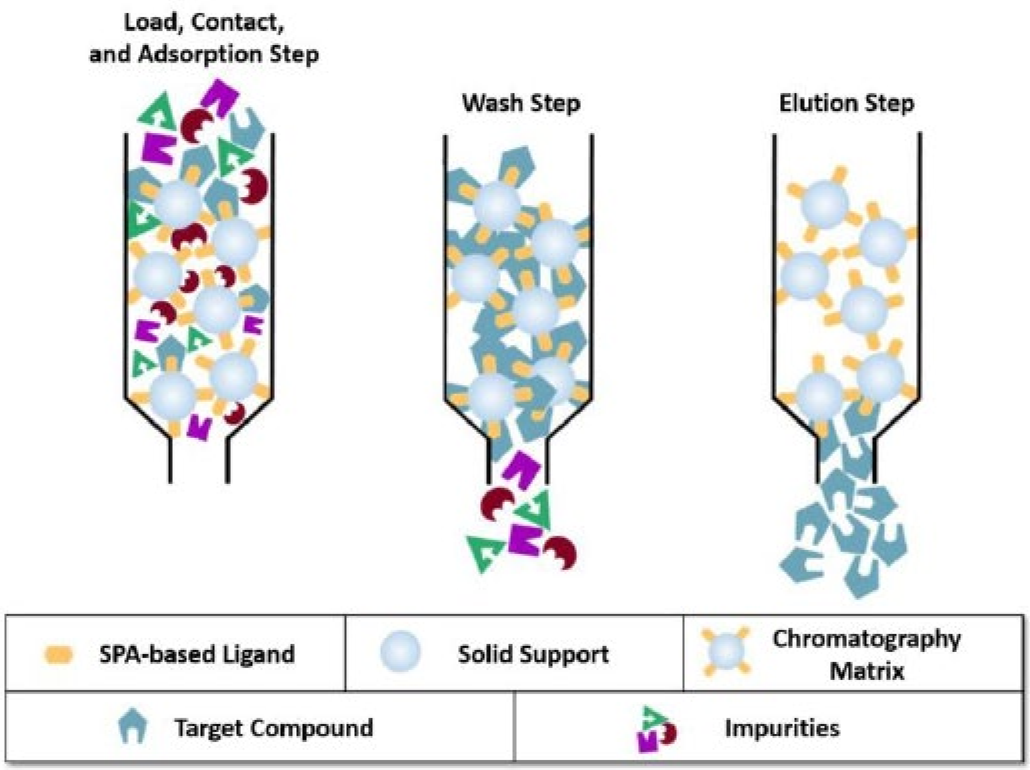


The figure (Ex. 2001, 2 (Fig. 1)), reproduced above, shows

> the structure of a conventional IgG and fragments that can be generated thereof. The constant heavy-chain domains C_H1, C_H2 and C_H3 are shown in yellow, the constant light-chain domain (C_L) in green and the variable heavy-chain (V_H) or light-chain (V_L) domains in red and orange, respectively. The antigen binding domains of a conventional antibody are Fabs and Fv fragments. Fab fragments can be generated by papain digestion. Fvs are the smallest fragments with an intact antigen-binding domain. They can be generated by enzymatic approaches or expression of the relevant gene fragments (the recombinant version). In the recombinant single-chain Fv fragment, the variable domains are joined by a peptide linker. Both possible configurations of the variable domains are shown, i.e. the carboxyl terminus of VH fused to the N-terminus of VL and vice versa.

Ex. 2001, 2; *see also* PO Resp. 5.

Below is a generic, exemplary schematic that shows how affinity purification typically works:



The figure shows the schematic of the loading, contact, and adsorbing step onto a column, followed by the wash step, and finally the elution and collection of the target compound. Ex. 1002 ¶ 24 (citing Ex. 1014 at §§ 1.1, 4.2.); *see also* PO Resp. 7 ("In a typical process, the composition containing the desired antibody then is loaded onto (i.e., pumped or injected into) the column."); Pet. 6; *see generally* Ex. 1014.

F. *The '765 patent (Ex. 1001)*

The '765 patent is titled "Chromatography Ligand Comprising Domain C from *Staphylococcus Aureus* Protein A for Antibody Isolation." Ex. 1001, (54). The '765 patent relates to an affinity ligand that is used for antibody isolation. *Id.* at 1:39–41. The '765 patent explains that chromatography is used in large-scale economic production of drugs and diagnostics in which proteins are produced by cell culture and then separated from the mixture of compounds and other cellular components to a sufficient purity. *Id.* at 1:52–61. One type of chromatography matrix for this purifying process includes immunoglobulin proteins, also known as antibodies, such as immunoglobulin G (IgG). *Id.* at 2:4–13. The '765 patent further explains that "[a]s in all process technology, an important aim is to keep the production costs low" by reusing matrices via cleaning protocols such as an alkaline protocol known as cleaning in place (CIP). *Id.* at 2:14–29. However, harsh treatments may impair the chromatography matrix materials such that there is a need for stability towards alkaline conditions for an engineered protein ligand. *Id.* at 2:31–48.

The '765 patent discloses that Protein A, known as SPA, is a constituent of the cell wall of the bacterium *Staphylococcus aureus*, and is widely used as a ligand in affinity chromatography matrices due to its ability to bind with IgG. *Id.* at 2:49–54. SPA is composed of five domains, designated in order from the N-terminus as E, D, A, B, and C, which are able to bind to antibodies at the Fc region, and it has been shown that each of these domains binds to certain antibodies at the Fab region. *Id.* at 2:54–63.

Domain C from SPA is defined by SEQ ID NO: 1 and is reproduced below.

```
Ala Asp Asn Lys Phe Asn Lys Glu Gln Gln Asn Ala Phe Tyr Glu Ile
1               5                   10                  15

Leu His Leu Pro Asn Leu Thr Glu Glu Gln Arg Asn Gly Phe Ile Gln
            20                  25                  30

Ser Leu Lys Asp Asp Pro Ser Val Ser Lys Glu Ile Leu Ala Glu Ala
        35                  40                  45

Lys Lys Leu Asn Asp Ala Gln Ala Pro Lys
    50                  55
```

Id. at 4:27; 15:1–20. SEQ ID NO: 1 shows Domain C has Glycine (Gly) as an amino acid at the position 29 as annotated via red highlighting. According to the ’765 patent, it has already been shown “that Domain C can act as a separate immunoglobulin adsorbent, not just as part of Protein A” and the ’765 patent discloses that from experiments, “the present inventors have quite surprisingly shown that the SPA Domain C presents a much improved alkaline-stability compared to a commercially available Protein A product.” *Id.* at 5:38–40, 51–55. The ’765 patent discloses, “it has been shown that an especially alkaline-sensitive deamidation rate is highly specific and conformation dependent, and that the shortest deamidation half times have been associated with the sequences -asparagine-glycine- and -asparagine-serine.” *Id.* at 5:62–66. The ’765 patent then discloses “[q]uite surprisingly, the Domain C ligand of the invention presents the herein presented advantageous alkaline-stability despite the presence of one asparagine-glycine linkage between residues 28 and 29” and “[t]hus, in a specific embodiment, the chromatography ligand according to the invention comprises SPA Domain C, as shown in SEQ ID NO 1, which in addition comprises the mutation G29A.” *Id.* at 5:67–6:3, 6:49–52. The ’765 patent discloses that a multimeric chromatography ligand (also denoted a “multimer”) can be comprised of at least two Domain C units and that a chromatography matrix can be comprised of ligands coupled to an insoluble carrier. *Id.* at 7:27–29, 8:21–23. The ’765 patent discloses a column study of alkaline stability of its Protein A-derived ligands and a testing of the Fab binding of its ligands. *Id.* at 10:32–14:59 (Example 1). The study includes using an injection liquid and solution along with human normal immunoglobulin as a target compound in chromatography experiments, ligand coupling and column packing, adsorbance measurements, washing out unbound samples, and eluting bound material. *Id.* at 11:11–18, 12:25–34, 13:32–37.

1. Illustrative Claim

Claims 1 and 14 are the independent claims challenged by Petitioner in this proceeding. Independent claim 1, reproduced below, is illustrative of the subject matter:

> 1. A chromatography matrix comprising:
>
> a solid support; and a ligand coupled to the solid support, the ligand comprising at least two polypeptides,
>
> wherein the amino acid sequence of each polypeptide comprises at least 55 contiguous amino acids of a modified SEQ ID NO. 1, and
>
> wherein the modified SEQ ID NO. 1 has an alanine (A) instead of glycine (G) at a position corresponding to position 29 of SEQ ID NO. 1.

Ex. 1001, 15:39–48. Claim 14 is similar to claim 1 but recites “at least 55 amino acids in alignment with SEQ ID NO. 1” rather than “at least 55 contiguous amino acids of a modified SEQ ID NO. 1,” and wherein “each polypeptide has an alanine (A) instead of glycine (G) at a position corresponding to position 29 of SEQ ID NO. 1” rather than “the modified

SEQ ID NO. 1 has an alanine (A) instead of glycine (G) at a position corresponding to position 29 of SEQ ID NO. 1." *Id.* at 16:51–60.

II. ANALYSIS

A. *Principles of Law*

"In an IPR, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review).

Petitioner must demonstrate by a preponderance of the evidence[4] that the claims are unpatentable. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d). A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time of the invention to a person having ordinary skill in the art. *KSR Int'l Co. v. Teleflex, Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including "the scope and content of the prior art"; "differences between the prior art and the claims at issue"; "the

[4] The burden of showing something by a preponderance of the evidence requires the trier of fact to believe that the existence of a fact is more probable than its nonexistence before the trier of fact may find in favor of the party who carries the burden. *Concrete Pipe & Prods. of Cal., Inc. v. Constr. Laborers Pension Tr. for S. Cal.*, 508 U.S. 602, 622 (1993).

level of ordinary skill in the art;" and "objective evidence of non-obviousness." *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art "to combine . . . known elements in the fashion claimed by the patent at issue." *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, "any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed." *Id.* at 420. Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that "a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so." *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (internal quotations and citations omitted).

B. *Level of Ordinary Skill in the Art*

In determining the level of skill in the art, we consider the "type of problems encountered in the art, [the] prior art solutions to those problems, [the] rapidity with which innovations are made, [the] sophistication of the technology, and [the] educational level of active workers in the field." *Custom Accessories, Inc. v. Jeffrey-Allan Indus. Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986); *Orthopedic Equip. Co. v. United States*, 702 F.2d 1005, 1011 (Fed. Cir. 1983).

Petitioner asserts that a person of ordinary skill in the art would have had

> (1) at least an advanced degree (*e.g.*, a Master's or Ph.D.) in biochemistry, process chemistry, protein chemistry, chemical engineering, molecular and structural biology, biochemical engineering, or similar disciplines; (2) several years of post-graduate training or related experience (including industry experience) in one or more of these areas; and (3) an understanding of the various factors involved in purifying proteins using chromatography.[] Such a person would have had multiple years of experience with affinity ligand design and protein purification.

Pet. 10–11 (citing Ex. 1002 ¶¶ 13–14). Patent Owner does not dispute Petitioner's definition of the person of ordinary skill. *See generally* PO Resp. Because Petitioner's proposed definition is unopposed and appears consistent with the Specification and art of record, we apply it here.

C. *Claim Construction*

The Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 42.200(b) (2021). Under that standard, claim terms "are generally given their ordinary and customary meaning" as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Petitioner argues that based on Patent Owner's implicit construction in the district court litigation "the term 'the ligand comprising at least two polypeptides' refers to a multimeric ligand (such as a tetramer) comprised of multiple polypeptides, each of which is a monomer." Pet. 17 (citing Ex. 1020 ¶¶ 41, 50, 58, 62, 74, 87).

Patent Owner does not contest Petitioner's construction. *See generally* PO Resp.

According to the Specification, "a multimeric chromatography ligand (also denoted a 'multimer') comprised of at least two Domain C units, or a functional fragments or variants thereof." Ex. 1001, 7:27–30. The Specification additionally recites that a multimer containing only Domain C units can, however, include linkers. *Id.* at 7:39–41. In addition, the Specification describes that "the multimer comprises one or more additional units, which are different from Domain C." *Id.* at 7:44–45. Based on these disclosures in the Specification, a multimer is composed of at least two or more monomers.

Because Petitioner's construction is consistent with the '765 patent's express construction of the term, and because Patent Owner does not contest Petitioner's construction, we apply it here.

D. *Overview of Asserted References*

1. Linhult (Ex. 1004)

Linhult is titled "Improving the Tolerance of a Protein A Analogue to Repeated Alkaline Exposures Using a Bypass Mutagenesis Approach." Ex. 1004, 1. Linhult discloses that due to the high affinity and selectivity of Staphylococcal protein A (SPA), "it has a widespread use as an affinity ligand for capture and purification of antibodies" but that "it is desirable to further improve the stability in order to enable an SPA-based affinity medium to withstand even longer exposure to the harsh conditions associated with cleaning-in-place (CIP) procedures." *Id.*, Abst. Linhult discloses, "[t]o further increase the alkaline tolerance of SPA, we chose to work with Z, which is a small protein derived from the B domain of SPA." *Id.* at 2.

Figures 1A and 1B of Linhult are reproduced below.

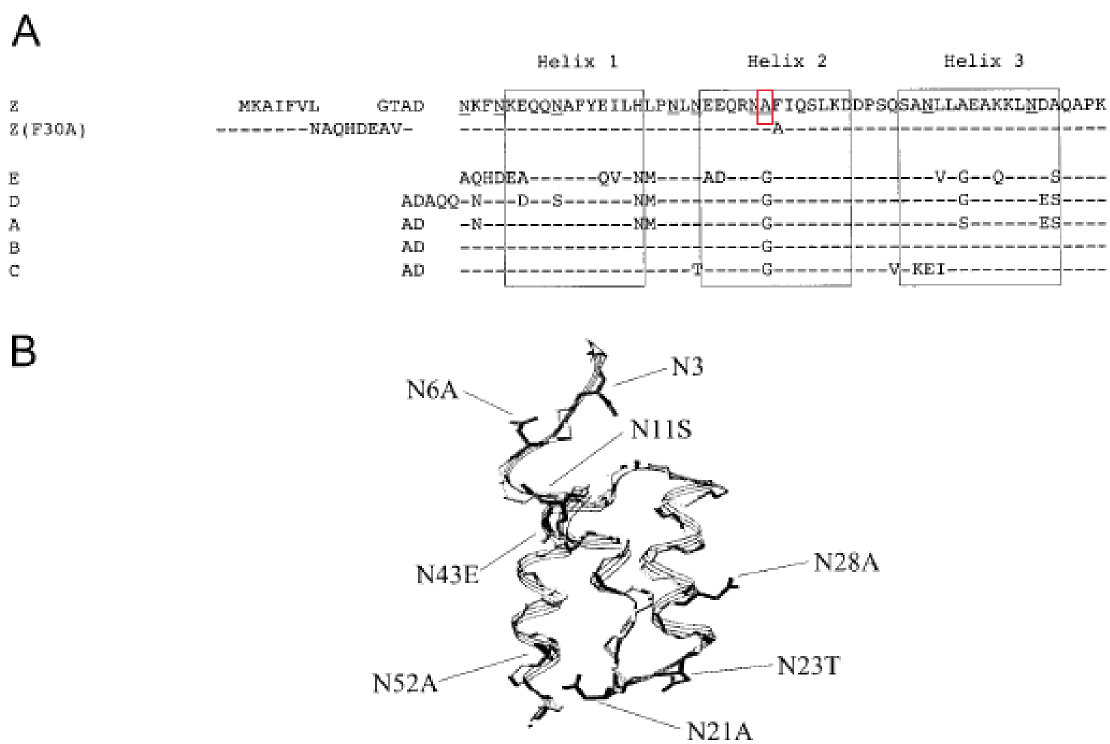


Figure 1A shows "[a]mino acid alignments of the Z, Z(F30A) and the five [naturally occurring] homologous domains (E, D, A, B, and C)" in which the horizontal lines indicate amino acid identity and "one glycine in the B domain [is] replaced [and] underlined" as annotated by the Board via a red box. *Id.* "Z(F30A), and all mutants thereof includes the same N-terminal as Z(F30A)" and "Z(N23T) was constructed with the same N-terminal as Z." *Id.*[5] Figure 1B shows "[t]he three-dimensional structure of the Z domain"

[5] The mutation N23T having a change in amino acid correlates with the amino acid N next to the "Helix 2" box of Figure 1A as annotated by Petitioner. *See* Pet. 13.

and "the different substitutions are indicated." *Id.* Specifically, Linhult discloses,

> [t]he B domain has been mutated in order to achieve a purification domain resistant to cleavage by hydroxylamine. An exchange of glycine 29 for an alanine has been made in order to avoid the amino acid combination asparagine–glycine, which is a cleavage site for hydroxylamine.[] Asparagine with a succeeding glycine has also been found to be the most sensitive amino acid sequence to alkaline conditions.[] Protein Z is well characterized and extensively used as both ligand and fusion partner in a variety of affinity chromatography systems.

Id. Using a 0.5 M NaOH cleaning agent and "a total exposure time of 7.5 h for Z(F30A) and mutants thereof," Linhult determines that "N23 seems to be very important for the functional stability after alkaline treatment of Z(F30A)" and "Z(F30A,N23T) shows only a 28% decrease in capacity despite the destabilizing F30A-mutation." *Id.* at 4–5; Figs. 2, 3. Linhult reports that "[h]ence, the Z(F30A,N23T) is almost as tolerant as Z and is thereby the most improved variant with Z(F30A) as scaffold." *Id.* at 5; Figs. 2, 3.

Linhult further discloses that "Z, Z(F30A), and mutated variants were covalently coupled to HiTrap™ affinity columns," that "[t]he Z domain includes 8 asparagines (N3, N6, N11, N21, N23, N28, N43, and N52; Fig. 1)," and that "since the amino acid is located outside the structured part of the domain, it will most likely be easily replaceable during a multimerization of the domain to achieve a protein A–like molecule." *Id.* at 4. Linhult confirms that "the affinity between Z(F30A) and IgG was retained despite the mutation." *Id.* In Linhult's studies, "[h]uman polyclonal IgG in TST was prepared and injected onto the columns in excess" and "[a] standard affinity chromatography protocol was followed." *Id.*

2. Abrahmsén (Ex. 1005)

Abrahmsén "relates to a recombinant DNA fragment coding for an immunoglobulin G ([]IgG) binding domain related to staphylococcal protein A . . . and to a process for cleavage of a fused protein expressed by using such fragment or sequence." Ex. 1005, 1:8–13. Abrahmsén discloses that "[b]y making a gene fusion to staphylococcal protein A any gene product can be purified as a fusion protein to protein A and can thus be purified in a single step using IgG affinity chromatography." *Id.* at 1:22–26. Abrahmsén explains that Protein A has "5 Asn-Gly in the IgG binding region of protein A" which "makes the second passage through the column irrelevant as the protein A pieces released from the cleavage will not bind to the IgG." *Id.* at 1:58–63. Abrahmsén provides a solution to this problem "by adapting an IgG binding domain so that no Met and optionally no Asn-Gly is present in the sequence." *Id.* at 1:64–67.

Abrahmsén discloses that in a preferred embodiment, "the glycine codon in the Asn-Gly constellation has been replaced by an alanine codon." *Id.* at 2:21–23. In one embodiment, Abrahmsén provides "a recombinant DNA sequence comprising at least two Z-fragments" in which "[t]he number of such amalgamated Z-fragments is preferably within the range 2–15, and particularly within the range 2–10." *Id.* at 2:27–31. Abrahmsén discloses that the recombinant DNA fragment can "cod[e] for any of the E D A B C domains of staphylococcal protein A, wherein the glycine codon(s) in the Asn-Gly coding constellation has been replaced by an alanine codon." *Id.* at 2:32–37. According to Abrahmsén, from a computer simulation of the Gly to Ala amino acid change, it was "concluded that this change would not interfere with folding to protein A or binding to IgG." *Id.* at 5:13–16.

3. Hober (Ex. 1006)

Hober "relates to . . . a mutant protein that exhibits improved stability compared to the parental molecule" and "also relates to an affinity separation matrix, wherein a mutant protein according to the invention is used as an affinity ligand." Ex. 1006, 1. Hober discloses that removal of contaminants from the separation matrix involves "a procedure known as cleaning-in-place (CIP)" but "[f]or many affinity chromatography matrices containing proteinaceous affinity ligands," the alkaline environment "is a very harsh condition and consequently results in decreased capacities owing to instability of the ligand." *Id.* at 1–2. According to Hober, structural modifications, such as deamidation and cleavage of the peptide backbone, of asparagine and glutamine residues in alkaline conditions is the main reason for loss of activity in alkaline solutions and that "the shortest deamidation half time have been associated with the sequences -asparagine-glycine and -asparagine-serine." *Id.* at 2. Further, from a study of a mutant of ABD that was created, it was concluded that "all four asparagine residues can be replaced without any significant effect on structure and function." *Id.* at 2–3. Hober points out that the SPA contains domains capable of binding to the Fc and Fab portions of IgG immunoglobulins from different species and reagents of this protein with their high affinity and selectivity have found a widespread use in the field of biotechnology. *Id.* at 3. Accordingly, "there is a need in this field to obtain protein ligands capable of binding immunoglobulins, especially via the Fc-fragments thereof, which are also tolerant to one or more cleaning procedures using alkaline agents." *Id.* at 4.

In one embodiment of Hober, a multimer "comprises one or more of the E, D, A, B, and C domains of Staphylococcal protein A" in which "asparagine residues located in loop regions have been mutated to more hydrolysis-stable amino acids" for advantageous structural stability reasons wherein "the glycine residue in position 29 of SEQ ID NO[]. 1 has also been mutated, preferably to, an alanine residue." *Id.* at 12. Hober's SEQ ID NO: 1 is reproduced below.

```
Ala Asp Asn Lys Phe Asn Lys Glu Gln Gln Asn Ala Phe Tyr Glu Ile
1               5                   10                  15

Leu His Leu Pro Asn Leu Asn Glu Glu Gln Arg Asn Gly Phe Ile Gln
            20                  25                  30

Ser Leu Lys Asp Asp Pro Ser Gln Ser Ala Asn Leu Leu Ala Glu Ala
        35                  40                  45

Lys Lys Leu Asn Asp Ala Gln Ala Pro Lys
    50                  55
```

Id. at SEQUENCE LISTING 1. SEQ ID NO: 1 shows a domain of Staphylococcus aureus having Glycine (Gly) as an amino acid at the position 29, as annotated via red highlighting.

Hober further discloses that its matrix for affinity separation "comprises ligands that comprise immunoglobulin-binding protein coupled to a solid support, in which protein at least one asparagine residue has been mutated to an amino acid other than glutamine." *Id.* at 13. For its method of isolating an immunoglobulin, Hober discloses "in a first step, a solution comprising the target compounds, . . . is passed over a separation matrix under conditions allowing adsorption of the target compound to ligands present on said matrix" and "[i]n a next step, a second solution denoted an eluent is passed over the matrix under conditions that provide desorption, i.e. release of the target compound." *Id.* at 16.

E. *Obviousness in view of Linhult, Abrahmsén, and Hober*

1. *Petitioner's Contentions*

a) *Claims 1 and 14*[6]

Petitioner argues that "*Linhult* explains that SPA-based chromatography matrices have 'widespread use in the field of biotechnology for affinity chromatography purification, as well as detection of antibodies.'" Pet. 19 (citing Ex. 1004, 1). Petitioner argues that Linhult teaches using a HiTrap™ chromatography affinity column made up of agarose beads that serve as a solid support for coupling SPA-based ligands. *Id.* (citing Ex. 1004, 4; Ex. 1002 ¶ 86). "Linhult discloses that its SPA-based ligands were 'coupled to' the solid support agarose beads [] contained in HiTrap™ affinity columns." *Id.* at 20 (citing Ex. 1004, 4). Petitioner argues that "Linhult discloses that 'multimerization' of SPA monomers is performed to 'achieve" an "[SPA-]like' affinity ligand." *Id.* (citing Ex. 1004, 4; Ex. 1002 ¶¶ 91–93). Petitioner argues that "Figure 1(a), *Linhult* describes at least 55 amino acids of SPA's naturally-occurring C domain (i.e., SEQ ID NO. 1)." *Id.* (citing Ex. 1004, 1, Fig. 1(a); *see* Ex. 1005, Fig. 2; Ex. 1006, Fig. 1; Ex. 1008, 639, Fig. 1). Petitioner argues "that all 'five SPA domains show individual affinity for the Fc-fragment . . . as well as certain Fab-fragments of [antibodies] from most mammalian species." *Id.* at 21 (citing Ex. 1004, 1). Petitioner argues that "*Abrahmsén* teaches a C(G29A)-based SPA ligand." *Id.* (citing Ex. 1002 ¶¶ 99–111).

[6] Petitioner treats claims 1 and 14 and their corresponding depend claims similarly; therefore, our analysis in this Decision groups the claims together.

Petitioner argues that it was "known that individual SPA domains, including the C domain, could be used to construct SPA-based affinity ligands for purifying proteins." *Id.* at 21 (citing Ex. 1002 ¶¶ 29, 101; Ex. 1004, 1; Ex. 1006, 12; Ex. 1018 ¶ 29; Ex. 1019, 6:25–34). Petitioner argues that "Linhult thus teaches a [person of ordinary skill in the art] that avoiding the Asn_{28}-Gly_{29} dipeptide sequence through a G29A mutation, including on the C domain, would yield an SPA-based ligand having increased alkali-stability." *Id.* at 22 (citing 1002 ¶¶ 100–04; Ex. 1011; Ex. 1012; Ex. 1013). Petitioner acknowledges that Linhult "does not expressly disclose a C(G29A)-based SPA ligand. Regardless, it would have been obvious to a [person of ordinary skill in the art] to modify *Linhult* based on the teachings of *Abrahmsén* to incorporate a C(G29A)-based SPA ligand in a chromatography matrix." *Id.* at 23 (Ex. 1002 ¶¶ 99–111). Petitioner argues "*Abrahmsén* expressly discloses 'a recombinant DNA coding for any of the E D A B C domains of [SPA], wherein the glycine codon(s) in the $Asn_{[28]}$-$Gly_{[29]}$ coding constellation has been replaced by an alanine codon.'" *Id.* at 23 (citing Ex. 1005, 2:32–37).

Petitioner concludes that

> applying the teachings of *Abrahmsén* with *Linhult* would have involved merely combining known elements in the field (e.g., an affinity chromatography matrix comprising a G29A-containing ligand coupled to a solid support, as in *Linhult*, and a C(G29A)-based amino acid sequence, as in *Abrahmsén*) according to known ligand-construction methods to yield a predictable results (e.g., the claimed affinity chromatography matrix).

Pet. 24.

Petitioner argues that Abrahmsén "unequivocally discloses including a G29A mutation on SPA's C domain." Pet. 49 (citing Ex. 1005, 2:32–37, 5:13–16; Ex. 1002 ¶¶ 222–27). The Petition recognizes that in Abrahmsén an IgG chromatography column was used to purify the recombinantly produced SPA. *Id.* In other words, the column matrix had IgG bound to the matrix and could capture the recombinantly produced SPA binding domains.

Petitioner relies on Hober for teaching that the substitution in G29A of SEQ ID NO: 1 would have been obvious and to address the conventional features set forth in dependent claims. *See* Pet. 20–26, 48–49. Specifically, Petitioner argues that Hober teaches "that naturally-occurring SPA 'contain[s] domains capable of binding to the Fc and Fab portions' of antibodies." Pet. 33 (citing Ex. 1006, 3); Ex. 1006, 3 ("An example of such a protein [used for affinity chromatography] is staphylococcal protein A, containing domains capable of binding to the Fc and Fab portions of IgG immunoglobulins from different species."). Petitioner argues that Hober also teaches that the shortest deamidation half lives are seen with sequences that contain an Asn-Gly dipeptide. Pet. 33 (citing Ex. 1005, 2); Ex. 1002 ¶ 76 ("*Hober* teaches that Asn-Gly dipeptide sequences are associated with 'the shortest deamidation half times,' which is a degradation process that occurs at high pH conditions."), *see id.* ¶ 141 (deamidation "is a process that degrades SPA-based ligands at high pH conditions.").

According to Petitioner, "*Hober* discloses, for example, that 'SPA-based affinity medium probably is the most widely-used affinity medium for isolation of monoclonal antibodies and their fragments from different samples.'" Pet. 50 (citing Ex. 1006, 3).

> *Hober* states, for example, that SPA-based monomers "can be combined into multimeric proteins, such as dimers, trimers, tetramers, pentamers etc." (Ex. 1006, 11; *see also* Ex. 1005, 9:14–10:41 (*Abrahmsén* disclosing Example V on "Construction

> of Dimeric Z Fragment”).) A [person of ordinary skill in the art] would have a good reason to combine the teaching from *Abrahmsén* with the teachings of *Hober* since multimeric SPA-based ligands allowed for additional surface area to bind target compounds. (Ex. 1002 ¶ 239.)

Pet. 52–53. Petitioner’s declarant, Dr. Cramer explains that using the ligand as a multimeric ligand has “the benefits of additional surface area on the SPA-based ligand to bind target compounds.” Ex. 1002 ¶ 239 (citing Ex. 1032, 560). Abrahmsén discloses that SPA domain C as shown in Figure 2 (*see* Ex. 1005 at col. 3:25-35, Fig. 2) has at least 55 contiguous amino acids of SEQ ID NO. 1. *See* Ex. 1002 ¶¶ 242–244. “*Abrahmsén* further confirms there are no structural integrity concerns in constructing a C(G29A)-based SPA ligand by disclosing that incorporation of a G29A mutation ‘would not interfere with folding [of SPA] or binding to [antibodies].’” Ex. 1002 ¶ 246.

b) Claims 4 and 17

With respect to claims 4 and 17, Petitioner argues that the property of being “capable” of binding to the Fab fragment of an antibody is an inherent property of SEQ ID NO:1 that has the glycine exchanged for the alanine at position 29. *See* Pet. 28. Specifically, Petitioner argues that “*Linhult* discloses that it was well known by 2004 that the five domains of SPA, including the C domain, ‘show individual ***affinity for***…certain ***Fab-fragments*** of [antibodies] from most mammalian species.’” Pet. 29 (citing Ex. 1004, 1; Ex. 1002 ¶ 125). “Even if the capability to bind the Fab part of an antibody is not an inherent property of the claimed C(G29A)-based SPA ligand, *Linhult* discloses this limitation.” *Id.* Petitioner notes further that “dependent claims 4 and 17 permit the incorporation of additional amino acid sequences known to possess the claimed property." *Id.* (citing Ex. 1002 ¶ 126).

c) Claims 2, 3, 12, 15, 16, and 25

With respect to claims 2, 3, 12, 15, 16, and 25, Petitioner directs our attention to where in the record the various limitations of the dependent claims may be found. *See* Pet. 25–28, 30. Specifically, Petitioner asserts that Linhult describes multimerization of the ligand as recited in claims 2, 12, 15, and 25. Pet. 26 (citing Ex. 1004, 410; Ex. 1002 ¶¶112–114), *see id.* at 30 (citing Ex. 1002 ¶¶ 128-129). Because the wild-type C-domain-based multimeric ligand inherently possesses the claimed binding-capacity property as recited in claims 3 and 16, there is a reasonable expectation that a chromatography possessing the C(G29A)-mutation would achieve similar binding capacity. Pet. 27–28 (citing Ex. 1002 ¶¶ 115–119).

2. Patent Owner's Contentions

Patent Owner argues that the Petition fails to demonstrate that it would have been obvious to make the chromatography matrix as claimed (PO Resp. 17–38); that the Petition has not established that there is a reasonable expectation of success in arriving at the claimed matrix (*id.* at 47–52); that the art teaches away from making the G29A modification (*id.* at 38–41); that the artisan would not have been motivated make additional mutations (*id.* at 43–44); and that objective indicia supports a conclusion of non-obviousness (*id.* at 53–53).

a) Matrix

According to Patent Owner, "Petitioners fail to explain why the POSA would have been motivated to select Domain C's amino acid sequence as the foundation for an engineered SPA ligand with favorable properties." PO Resp. 18. Specifically arguing that the obviousness analysis requires the prior art be viewed as a whole. PO Resp. 20 (citing *In re Wesslau*, 353 F.2d 238 (CCPA 1965); *In re Enhanced Sec. Rsch., LLC*, 739 F.3d 1347, 1355 (Fed. Cir. 2014); *Impax Lab'ys Inc. v. Lannett Holdings Inc.*, 893 F.3d 1372, 1379 (Fed. Cir. 2018)).

Patent Owner argues that because nobody was working on Domain C at the time the invention was filed, therefore, the selection of Domain C for further development could not possibly be obvious. *See* PO Resp. 23 ("Reliance on *KSR* also is foreclosed by the evidence that no one in the art was seeking to modify Domain C."), *see also id.* at 24 ("But no prior art cited by Petitioners singles Domain C out for further development. Ex. 2025 ¶¶ 89-96"), *id.* at 26 ("Dr. Cramer [Petitioner's expert] himself highlights, it would have been natural for the POSA to further develop the domain—Domain B—that was best understood and for which there was a crystal structure available. Ex. 1002 ¶ 33; Ex. 2015 at 137:20–138:19; Ex. 2017"), *id.* at 28 ("The notion that this body of work would lead the POSA to discard the improved ligands the references themselves focus on, and instead start experimenting with mutations to Domain C—strains credulity. Ex. 2025 ¶¶ 92–95").

According to Patent Owner, neither Linhult nor Abrahmsén supply the motivation to start with Domain C. "Linhult focuses exclusively on, and concerns improvements to, the alkaline stability of Domain Z by mutating asparagine residues. *See* Ex. 2025 ¶¶ 64-67, 92." PO Resp. 29. "Rather than use Domain C, the POSA reviewing Linhult would be motivated to keep working with Domain Z, adopting the N23T mutation. Ex. 2025 ¶ 67 & n.3." PO Resp. 30. "Neither Abrahmsén itself nor the Petition provide any reason as to why the POSA would have 'plucked' Domain C from among the five listed SPA domains. *WBIP*[*LLC v. Kohler Co.*, 829 F.3d 1317, 1337 (Fed. Cir. 2016)]." PO Resp. 31.

b) Reasonable Expectation of Success

Patent Owner argues that "the field of protein engineering is notoriously unpredictable." PO Resp. 21 (citing Ex. 2025 ¶¶ 50-52). Arguing that "despite their supposed structural similarity, there are a number of differences between the naturally-occurring domains of protein A, including five different amino acids in the sequences of Domain B (with which the industry was quite familiar) and Domain C (which remained virtually ignored as of the priority date)." *Id.* at 22 (citing Ex. 2025 ¶ 48).

> Protein engineering is a highly complex and unpredictable field and was all the more so as of the priority date more than fifteen years ago. *See*, *e.g.*, Ex. 2025 ¶¶ 50-52. . . . As amply demonstrated by the effect of the G29A mutation on Domain Z's Fab-binding ability, even a single amino acid substitution can drastically alter the properties of a protein. Ex. 2025 ¶ 52; Ex. 2015 at 51:15-52:1 ([Dr. Cramer, Petitioner's expert] agreeing that a single amino acid change can have a significant effect on a ligand's binding ability), 18:10-12, 73:16-20.

PO Resp. 34–35.

Patent Owner argues that

> The Federal Circuit has rejected arguments premised on the notion that a homologous structure renders an invention obvious, particularly given the difficulty and uncertainty in the art as of

> the priority date. *See, e.g., Amgen, Inc. v. Chugai Pharm. Co*., 927 F.2d 1200, 1208 (Fed. Cir. 1991) (holding the use of a monkey gene to probe for a roughly 90 percent "homologous" human gene would not have been obvious, particularly given expert testimony that isolating a particular gene would have been "difficult" and the lack of certainty in the endeavor).

PO Resp. 36. Specifically, Patent Owner argues that the Fab-binding capability of a ligand could not have been predicted and therefore there is no reasonable expectation of success. *See* PO Resp. 38.

c) Teaching Away

Patent Owner argues that the prior art would have told the person of ordinary skill in the art to avoid a G29A a modification to Domain C. PO Resp. 38. In other words, its Patent Owner's contention is that the prior art teaches away from making this modification. "The very G29A amino acid substitution Petitioners now suggest the POSA would seek to employ with Domain C would have been known to have rendered Fab binding 'negligible' when implemented in Domain B." PO Resp. 40. Patent Owner argues that a person seeking to improve Fab binding one avoid a G29A substitution of Domain C. PO Resp. 40–41.

d) Additional Modifications

Patent Owner argues that "the prior art would have taught the POSA to make asparagine substitutions, not glycine substitutions, to address alkaline stability concerns." PO Resp. 43. In other words, Patent Owner's argues the prior art would have suggested making additional substitutions most notably in the asparagine residues of Domain C. *Id.* at 44.

e) Inherency

Patent Owner argues that Petitioner cannot rely on inherency to circumvent that the combination lacks a reasonable expectation of success. PO Resp. 52 (citing *Honeywell Int'l Inc. v. Mexichem Amanco Holding S.A. DE C.V.*, 865 F.3d 1348, 1355 (Fed. Cir. 2017)). "As Dr. Bracewell [Patent Owner's expert] explains, the POSA did not know and could not have known the properties of a modified Domain C ligand, since the effects of even single-amino acid substitutions are undisputedly unpredictable and the cited art includes absolutely no protein engineering work with Domain C. Ex. 2025 ¶¶ 120-122." PO Resp. 48. According to Patent Owner, because nobody was working on domain C the person of ordinary skill in the art "is simply left to guess at how such a ligand would perform." PO Resp. 49 (citing Ex. 2025 ¶¶ 121–124). Specifically, the person of ordinary skill in the art would not know whether a Domain C-G29A mutation would be capable of binding Fab part of an antibody. *Id.*

f) Unexpected Results

Patent Owner argues that it was wholly unexpected that the modified C(G29A)-based SPA ligand to bind the Fab part. PO Resp. 55. "[T]he SPA ligands of the claimed chromatography matrices unexpectedly retained their ability to bind to the Fab part of an antibody despite the substitution of an alanine for the glycine at position 29 of the Domain C sequence." PO Resp. 55.

3. Petitioner's Reply

In response, Petitioner argues that

> *Abrahmsén* and *Hober* each expressly pointed to a C(G29A) mutation (Ex. 1005, 2:32-37; Ex. 1006, 12), which was known to increase alkali-stability by avoiding the troublesome Asn_{28}-Gly_{29} dipeptide sequence (*see, e.g.*, Ex. 1004, 408). As the Board recognized, "*Abrahmsén* provides motivation for making [the G29A] mutation in ***any of the IgG binding domains***." (Decision, 24; *see also* Ex. 1057, 97:3-16 (Dr. Bracewell admitting that *Abrahmsén* discloses a G29A mutation to any of the five domains, including Domain C).)

Reply 2.

> A POSA would have reasonably expected success in combining these teachings to achieve the claimed affinity chromatography matrix given the well-known fact that each individual domain, including Domain C, has affinity for antibodies (Ex. 1004, 407), as well as *Abrahmsén*'s confirmation that G29A "would not interfere with folding [of SPA] or binding to [antibodies]" (Ex. 1057, 99:13-101:21 Ex. 1005, 5:13-16; Ex. 1002 ¶110).

Id. at 3.

Petitioner argues that Patent Owner "has not disputed that *Abrahmsén* disclosed that G29A 'would not interfere with folding to protein A or binding to IgG.' (Ex. 1005, 2:32-37, 5:4-16; Ex. 1057, 109:20-110:17.) Nor does it take issue with its own statements in *Hober* that G29A is advantageous for 'structural stability reasons.' (Ex. 1006, 12.)" *Id.* at 9. "*Abrahmsén* and *Hober*, which make clear that G29A does not affect the ability of an SPA ligand to bind to an antibody. (Ex. 1005, 2:32-37, 5:4-16; Ex. 1006, 12.)" *Id.* at 10.

Petitioner argues that "a POSA would have started with any one of the naturally occurring domains. (Decision, 26-28.) To then increase alkali stability, a POSA would have made the simplest, well-known substitution: G29A. (Section II.A.1-2; Ex. 1061 ¶¶8-15.)" Reply 11.

Petitioner argues "that Fab-binding was an inherent feature of a C(G29A)-based ligand—which [Patent Owner] does not appear to dispute. (Decision, 30; [PO] Resp., 52-53.) In fact, [Patent Owner] acknowledges that 'C(G29A)-based SPA ligands retained substantial Fab-binding ability.' (Resp., 55.)" *Id.* at 14.

Petitioner argues that "Fab-binding is not being used [in the Petition] as part of a finding of a motivation to combine; rather, it is an inherent property [of the composition] being claimed. And necessarily present properties do not add patentable weight when they are claimed as limitations. *In re Kubin*, 561 F.3d 1351, 1357 (Fed. Cir. 2009)." Reply 15–16. Petitioner further argues that Patent Owner's reliance on *Honeywell* is misplaced because "*Honeywell* had to do with an inherent property being used as a teaching in an obviousness analysis; it did not involve a limitation in the challenged claim reciting an inherent property." Reply 15 (citing *Honeywell Int'l Inc. v. Mexichem Amanco Holding S.A. De C.V.*, 865 F.3d 1348, 1355 (Fed. Cir. 2017); *see also Pernix Ireland Pain v. Alvogen Malta Operations*, 323 F. Supp. 3d 566, 607(D. Del. 2018)).

4. *Patent Owner's Sur-reply*

Patent Owner argues that "Petitioners, and the Institution Decision, overlook an important point of consensus between the parties' experts: the field of protein engineering is notoriously *unpredictable*." Sur-reply 2. Patent Owner maintains that Petitioner has not identified a motivation to start from Domain C. *Id.* at 3. Patent Owner argues that "Petitioners would have the Board look past the multitude of references teaching a preference for Domains B and Z—including Petitioners' foundational references—and seize upon fleeting mentions of Domain C." *Id.* at 7 (citing *In re Wesslau*, 353 F.2d 238, 241 (C.C.P.A. 1965)"). Patent Owner argues that "[m]ere sequence homology does not make the field predictable, as both experts observe, Ex. 2015 at 51:15-52:1, 56:4-12, 75:12-22, 73:16-20; Ex. 2025 ¶¶ 50-52; Ex. 2049 at 72:1-73:12, and as the vastly different Fab-binding properties of the near-identical Domains B and Z well illustrate, Ex. 2029 at 8." *Id.* at 8–9.

Patent Owner argues that "Abrahmsén's computer simulation was of unmodified Protein A as a whole, not a Domain C (or G29A-modified) monomer or multimer, and thus does not reveal the impact of a G29A mutation on protein folding or IgG affinity. Ex. 2025 ¶ 103; Ex. 2049 at 131:7-10." Sur-reply 11.

5. Analysis

a) Claims 1 and 14

Claims 1 and 14 of the '765 patent are directed to a composition. Specifically, a chromatography matrix (i.e. a solid support) that has a ligand attached, and that ligand is made up of at least two polypeptides comprising 55 contiguous amino acids of SEQ ID NO: 1[7] having a G29A mutation. *See* Ex. 1001, 15:39–48.

Linhult teaches that SPA is a cell surface protein expressed by *Staphylococcus aureus* and consists of five highly homologous domains (EDABC). Ex. 1004, 1. Each of "[t]he five SPA domains show individual affinity for the Fc-fragment [11 residues of helices 1 and 2 (domain B)], as

[7] Wild type amino acid sequence of domain C from *Staphylococcus* protein A (SPA). *See* Ex. 1001, 4:27, 6:35–36, 6:51–52.

well as certain Fab-fragments of immunoglobulin G (IgG) from most mammalian species." *Id.* "Due to the high affinity and selectivity of SPA, it has a widespread use as an affinity ligand for capture and purification of antibodies." Ex. 1004, Abst., *see also id.* at 1 ("SPA has a widespread use in the field of biotechnology for affinity chromatography purification, as well as detection of antibodies.").

Linhult explains that, in column chromatography, sodium hydroxide (NaOH) is probably the most extensively used cleaning agent for removing contaminants such as nucleic acids, lipids, proteins, and microbes, and a cleaning-in-place (CIP) step is often integrated in the protein purification protocols using chromatography columns. Ex. 1004, 1. "Unfortunately, protein-based affinity media show high fragility in this extremely harsh environment, making them less attractive in industrial-scale protein purification. SPA, however, is considered relatively stable in alkaline conditions." *Id.* at 2. Linhult teaches that the combination of "[a]sparagine with a succeeding glycine has also been found to be the most sensitive amino acid sequence to alkaline conditions." *Id.* Linhult teaches that "[a]n exchange of glycine 29 for an alanine has been made in order to avoid the amino acid combination asparagine–glycine, which is [also] a cleavage site for hydroxylamine." *Id.*

Petitioner's declarant, Dr. Cramer explains that the "Z" domain referenced in Linhult refers to a synthetic version of the wild-type (i.e., natural) B domain of SPA, in which the naturally occurring glycine in the Asn_{28}-Gly_{29} dipeptide sequence is replaced by an alanine residue to create an Asn_{28}-Ala_{29} dipeptide sequence. Ex. 1002 ¶ 30 (citing Ex. 1004, 408, Fig. 1(a); Ex. 1007, 109, Fig. 1); ¶ 31 (citing Ex. 1005).

We credit Dr. Cramer for establishing that the C domain sequence disclosed in Linhult contains 55 amino acids of domain C as recited in SEQ ID NO: 1. Ex. 1002 ¶ 97 (showing a sequence alignment), *see also id.* ¶ 243 (showing sequence alignment of domain C of Abrahmsén with SEQ ID NO: 1); *see also* Ex. 2025 ¶ 41 (Patent Owner's declarant Dr. Bracewell showing SPA sequence alignment for Domains A, B, C, D, E, and Z).

Linhult teaches making affinity chromatography columns with protein Z, Z(F30A), and additional mutated variants. These modified proteins were covalently attached to HiTrap™ columns in Linhult using NHS-chemistry. Ex. 1004, 4. Human polyclonal IgG was prepared and injected onto the columns in excess and "[a] standard affinity chromatography protocol was followed." *Id.* at 4. Linhult exemplifies using the Z domain and Z domain mutants attached to the column for the isolation of IgG from a sample. *Id.* Linhult, therefore, teaches column chromatography matrix that has a SPA domain containing a G29A mutation attached to a chromatography matrix.

According to Abrahmsén, the IgG binding domains of E D A B C domains of SPA were known. *See* Ex. 1005, 3:25–35, 4:34–37, Fig. 2. Abrahmsén teaches that the Asn-Gly dipeptide is located in the middle of an alpha helix involved in the binding to IgG. *Id.* 4:56–60; Ex. 2025 ¶ 41 (showing alpha helix regions in SPA domains). Relying on "computer analysis [Abrahmsén] surprisingly showed that the Gly in the Asn-Gly dipeptide sequence could be changed to an Ala. This change was not obvious as glycines are among the most conserved amino acids between homologous protein sequences due to their special features." Ex. 1005, 5:7–9. Abrahmsén teaches that "the glycine codon in the Asn-Gly constellation has been replaced by an alanine codon." *Id.* at 2:21–23.

Abrahmsén teaches "a recombinant DNA fragment coding for *any of the E D A B C domains* of Staphylococcal protein A, wherein the *glycine codon(s) in the Asn-Gly coding constellation has been replaced by an alanine codon*." *Id.* at 2:33–37 (emphasis added). Abrahmsén, therefore, provides motivation for making this mutation in *any one* of the IgG binding domains of E D A B C domains of SPA.

Abrahmsén, like Linhult, only exemplifies the cloning and expression of the Z-domain, which is a B-domain with a G29A mutation. Ex. 1005, 7:65–10:56. A references disclosure, however, is not limited only to its preferred embodiments, but is available for all that it discloses and suggests to one of ordinary skill in the art. *In re Lamberti*, 545 F.2d 747, 750 (CCPA 1976); *see also In re Susi*, 440 F.2d 442, 446 n.3 (CCPA 1971) (disclosed examples and preferred embodiments do not constitute a teaching away from a broader disclosure or non-preferred embodiments). Here, Abrahmsén expressly suggests making the same mutation in *any one* of the SPA domains E D A B C. Ex. 1005, 2:33–37

Thus, the disclosures of both Linhult and Abrahmsén suggest mutating the glycine at position 29 for an alanine in an any one of the SPA IgG binding domains of E D A B or C in order to avoid protein degradation in alkaline conditions and degradation by hydroxylamine. *See* Ex. 1004, 2 ("An exchange of glycine 29 for an alanine has been made in order to avoid the amino acid combination asparagine–glycine, which is a cleavage site for hydroxylamine. Asparagine with a succeeding glycine has also been found to be the most sensitive amino acid sequence to alkaline conditions."); Ex. 1005, 4:56–57 ("The Asn-Gly dipeptide sequence is sensitive to hydroxylamine."); Ex. 1006, 2 ("and the shortest deamidation half times have been associated with the sequences –asparagine–glycine and –asparagine–serine”); Reply 3–5; Ex. 1061 ¶¶ 3–6.

Based on these disclosures, the combined teachings of Linhult and Abrahmsén suggest making the G29A mutation in *any one* of the SPA IgG binding domains E, D, A, B, or C. We, therefore, agree with Petitioner that attaching any other mutated SPA IgG binding domains E, D, A, or C using “known ligand-construction methods to yield a predictable result[] (e.g., the claimed affinity chromatography matrix)” would have been obvious. Pet. 24 (citing Ex. 1002 ¶ 109). As the Federal Circuit has explained, “[w]here a skilled artisan merely pursues ‘known options’ from ‘a finite number of identified, predictable solutions,’ the resulting invention is obvious under Section 103.” *In re Cyclobenzaprine Hydrochloride Extended-Release Capsule Patent Litig.*, 676 F.3d 1063, 1070 (Fed. Cir. 2012) (quoting *KSR*, 550 U.S. at 421).

Accordingly, we agree with Petitioner that the combination of Linhult and Abrahmsén expressly suggests mutating the glycine codon for an alanine codon in *any one* of the SPA IgG binding domains E, D, A, B, or C. Pet. 23 (citing Ex. 1002 ¶¶ 99–111).

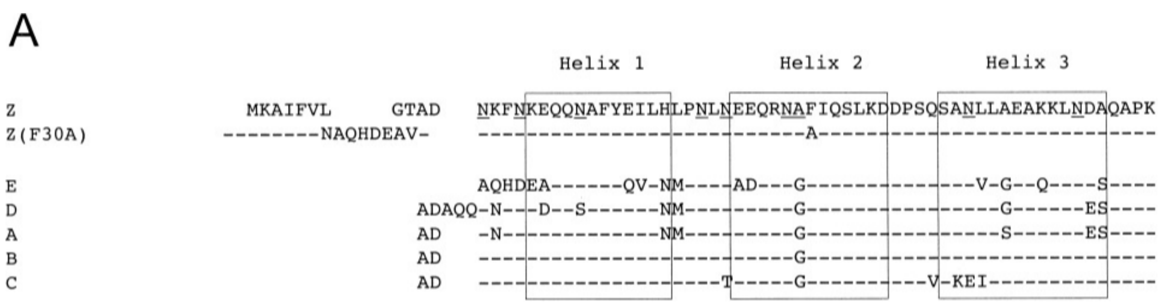
We address Patent Owner’s contentions below.

(1) Matrix

We do not find Patent Owner’s argument that the Petition fails to identify a reason to select domain C persuasive. PO Resp. 17–30. Specifically, we are not persuaded by Patent Owner’s contention that just because nobody was working on Domain C at the time the invention was filed, therefore, selection of Domain C cannot be obvious. *See* PO Resp. 31

("A general recognition that there exist five naturally occurring protein A domains is not a motivation to use each of them as a starting point for the claimed mutations").

Petitioner's articulated obviousness ground is premised on the knowledge that *any one* of the five SPA IgG binding domains are known to bind IgG and can function as a ligand for the purification of antibodies. Linhult and Abrahmsén both expressly suggest that the glycine codon at position 29 can be mutated for an alanine codon in *any one* of the SPA IgG binding domains E, D, A, B, or C. Ex. 1004, 2; Ex. 1005, 2:32–37. The SPA IgG binding domains comprise a short list of 5 members: E, D, A, B, or C. Of these 5 members, the glycine at position 29 in Domain B has already been mutated to an alanine to create a Domain Z which has been shown to retain IgG binding activity. Ex. 1004, 6 (Fig. 3). Figure 1A of Linhult is reproduced below.



Linhult's Figure 1A, reproduced above, shows the amino acid alignments of the Z, Z(F30A) and the five homologous domains (E, D, A, B, and C). The three boxes show the α-helices. Ex. 1004, 2; Ex. 1005, Fig. 2.

As discussed in our Institution Decision (Dec. 26–27), "it is fair to say that there were 'a finite number of identified, predictable solutions' to the problem of finding" a SPA IgG binding domain that is resistant to protein degradation by mutating the glycine at position 29 for an alanine and this is a "product not of innovation but of ordinary skill and common sense." S*ee Wm. Wrigley Jr. Co. v. Cadbury Adams USA LLC*, 683 F.3d 1356, 1364-65 (Fed. Cir. 2012) (quoting *KSR*, 550 U.S. at 421). It is well established that

> [s]tructural relationships may provide the requisite motivation or suggestion to modify known compounds to obtain new compounds. For example, a prior art compound may suggest its homologs because homologs often have similar properties and therefore chemists of ordinary skill would ordinarily contemplate making them to try to obtain compounds with improved properties.

In re Deuel, 51 F.3d 1552, 1558 (Fed. Cir. 1995). Here, Linhult and Abrahmsén show that the IgG binding domains of SPA – E, D, A, B, or C share many structural similarities. *See* Ex. 1004, 2 (Fig. 1(a) (reproduced above)); Ex. 1005, 3:25–35.

There is also an express teaching in both Linhult and Abrahmsén to mutate the glycine at position 29 to an alanine in order to prevent degradation of the protein and increase stability, which supports the obviousness of incorporating the mutation into any IgG binding domain that has the Asn-Gly dipeptide. *See, e.g.*, *SIBIA Neurosciences, Inc. v. Cadus Pharm. Corp.*, 225 F.3d 1349, 1358–59 (Fed. Cir. 2000) (stating that an express teaching in the prior art suggesting a particular modification establishes obviousness).

Accordingly, we find that Petitioner has shown by a preponderance of the evidence that the combined teachings of Linhult and Abrahmsén suggests the use of *any one* of the SPA IgG binding domains E, D, A, B, or C as the starting ligand for purifying IgG antibodies, and that making the G29A mutation in *any one* of the domains would have been obvious because it would provide ligands that are less susceptible to alkaline conditions and are resistant to hydroxylamine cleavage. Pet. 21–25; Reply 3–5; Ex. 1061 ¶¶ 8–11; Ex. 1004, 2; Ex. 1005, 2:32–37.

(2) Reasonable Expectation of Success

We are not persuaded by Patent Owner's contention that there is no reasonable expectation of success in using a G29A mutation in Domain C. PO Resp. 39–40; 47–53.

Linhult explains that removing the asparagine–glycine amino acid combination not only results in the removal of the hydroxylamine cleavage site but also creates a product that is more alkaline resistant. *See* Ex. 1004, 2 ("An exchange of glycine 29 for an alanine has been made in order to avoid the amino acid combination asparagine–glycine, which is a cleavage site for hydroxylamine. Asparagine with a succeeding glycine has also been found to be the most sensitive amino acid sequence to alkaline conditions.").

Abrahmsén teaches that this Asn-Gly amino acid combination is present in all five IgG biding domains and that mutating the dipeptide would not interfere with IgG biding. Ex. 1005, 4:56–58 ("The Asn-Gly dipeptide sequence is sensitive to hydroxylamine. As this sequence is kept intact in all five IgG binding domains of protein A. . . . However, by simulating the Gly to Ala amino acid change in the computer we concluded that this change would not interfere with folding to protein A or binding to IgG."). Abrahmsén's conclusion that the mutation would not interfere with biding to IgG is supported by Linhult (*see* Ex. 1004, 6 (Fig. 3)) and Jansson.[8]

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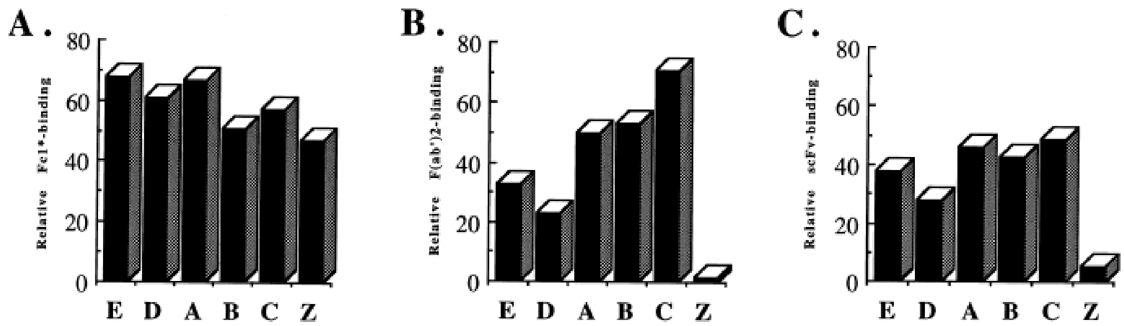
[8] Patent Owner cites Jansson for the position that Domain Z has negligible binding to Fab. *See* PO Resp. 39–40. Claims 1 and 14 are not limited to Fab binding, indeed the claims do not even require IgG binding.

Jansson's Fig. 3, reproduced below, shows a side-by-side comparison of Fc1*, Fab, and scFv binding.



Jansson Figure 3 (Panel A), reproduced above, shows that the single G29A mutation between Domain B and Domain Z results in a protein that is able to bind IgG.[9] Comparing panel A–B column with panel A–Z column, the relative binding capacity in both columns remains close to 50% for both, indicating that the G29A mutation does not interfere with IgG binding. Ex. 2009, 6. This is a result already predicted by Abrahmsén's computer modeling and substantiated by Linhult. *See* Ex. 1005, 4:56–58; Ex. 1004, 6 (Fig. 3).

"Obviousness does not require absolute predictability of success . . . all that is required is a reasonable expectation of success." *In re Droge*, 695 F.3d 1334, 1338 (Fed. Cir. 2012) (quoting *In re Kubin*, 561 F.3d 1351, 1360 (Fed. Cir. 2009) (citing *In re O'Farrell*, 853 F.2d 894, 903–04

[9] Fc1* is the constant region of human IgG1. Ex. 2029, 4. Fc1* is understood to be used as the "IgG control" in Jansson. Patent Owner's counsel explains that "Part A is Fc binding. So that is, I believe the way they did this experiment was with Fc fragments, but it's generally acknowledged, you know, these antibodies all have an Fc domain if they're a whole antibody and that reflects the fact that all of these domains A, B, C, D and E and domain Z, which is B with the G29A mutation, retain this Fc binding." Tr. 70:6–11.

(Fed.Cir.1988)); *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359 (Fed. Cir. 2016) (explaining that the expectation of success issue involves a showing of "a reasonable expectation of achieving *what is claimed*") (emphasis added). It is not inventive to confirm something that was already suggested in the art. "Scientific confirmation of what was already believed to be true may be a valuable contribution, but it does not give rise to a patentable invention." *Pharma Stem Therpeutic, Inc. v. ViaCell, Inc.,* 491 F.3d 1342, 1363–1364 (2007).

Here, the record supports that each individual SPA domain, including the C domain, has affinity for IgG antibodies. Ex. 1004, 1 ("The five SPA domains show individual affinity for the Fc-fragment [11 residues of helices 1 and 2 (domain B)], as well as certain Fab-fragments of immunoglobulin G (IgG) from most mammalian species." (bracketing in original)). Abrahmsén suggests making the mutation of Asn-Gly coding constellation in *any one* of the SPA domains by replacing glycine with an alanine codon that would remove the dipeptide sequence known to be sensitive to hydroxylamine degradation. *See* Ex. 1005, 4:56–5:16, *see also id.* Fig. 2 (showing the Asn-Gly coding constellation in all SPA domains); Ex. 1006, 2 ("the shortest deamidation half times have been associated with the sequences – asparagine–glycine and – asparagine–serine"). Abrahmsén's confirms that a G29A mutation on SPA would not interfere with folding of SPA protein and the binding to antibodies. Ex. 1005, 5:13–16 ("by simulating the Gly to Ala amino acid change in the computer we concluded that this change would not interfere with folding to protein A or binding to IgG."). Abrahmsén's computer modeling suggests that IgG binding is not impacted by the mutation and this is confirmed by Linhult's experiments showing that the

G29A mutant of Domain B (a.k.a. Domain Z) binds IgG. Ex. 1004, 6 (Fig. 3).

Patent Owner argues that "Abrahmsén's computer simulation was of unmodified Protein A as a whole, not a Domain C (or G29A-modified) monomer or multimer, and thus does not reveal the impact of a G29A mutation on protein folding or IgG affinity. Ex. 2025 ¶ 103; Ex. 2049 at 131:7-10." Sur-reply 11.

We are not persuaded by Patent Owner's contention that the information gained by computer modeling of the SPA native domain B – IgG crystal structure could not be extrapolated to other SPA domains that are structurally very similar.

As Petitioner's expert, Dr. Cramer explains

> It was well known that the researchers who developed the Z domain based on the wild-type B domain (rather than any of the other four SPA domains) did so for two reasons. (*See*, *e.g.*, Ex. 1007 at 109.) First, a crystal structure of the wild-type B domain binding to an antibody happened to be available in 1981 for analysis. (*See*, *e.g.*, *id.*; Ex. 1005 at col. 4:56-68; Ex. 1017.) And, second*, their work would be informative of mutations that could be done on all five of the highly homologous SPA domains more generally*. (*See*, *e.g.*, Ex. 1005 at col. 2:32-37; Ex. 1007 at 109; Ex. 1008 at 639, Fig. 1.)

Ex. 1002 ¶ 33 (emphasis added). Dr. Cramer further explains that "[t]hey did the computer modeling based on that complex because that's the crystal structure that they had. It wasn't done because the B domain is special. . . . And then there's several other places where they state clearly that they could also do the other domains with expected similar results." Ex. 2015, 138:8–22.

Accordingly, we agree with Petitioner that taken together, the teachings of Linhult, Abrahmsén, and Hober provide a reasonable expectation of success at arriving at a chromatography composition that contains the SPA domain C ligand with the G29A mutation. Pet. 25 (citing Ex. 1005, 5:13–16; Ex. 1002 ¶ 110).

(3) No Teaching Away

We are also not persuaded by Patent Owner's contention that the art teaches away from the G29A substitution because it interferes with Fab binding. *See* PO Resp. 38–40; Sur-reply 9; Ex. 2009 at 2; Ex. 2010 at 25; Ex. 2012 at 25–26; Ex. 2029 at 7.

Neither claim 1 nor claim 14 recite a need to bind the Fab region of an antibody – all that is required by these claims is a chromatography matrix with a domain C ligand attached, and the domain C sequence having a G29A mutation.[10] The law does not require that the teachings of the reference be combined for the reason or advantage contemplated by the inventor, as long as some suggestion to combine the elements is provided by the prior art as a whole. *In re Beattie*, 974 F.2d 1309, 1312 (Fed. Cir. 1992); *In re Kronig*, 539 F.2d 1300, 1304 (CCPA 1976); s*ee In re Kemps*, 97 F.3d 1427, 1430 (Fed. Cir. 1996) ("[T]he motivation in the prior art to combine the references does not have to be identical to that of the applicant to establish obviousness.").

Here, Linhult teaches that "[t]he five SPA domains show individual affinity for the Fc-fragment [11 residues of helices 1 and 2 (domain B)], as well as certain Fab-fragments of immunoglobulin G (IgG) from most

[10] We note that neither claim 1 nor claim 14 even requires IgG binding.

mammalian species." Ex. 1004, 1 (bracketing in original) (citation omitted). Linhult, therefore, teaches that *any one* of the SPA IgG binding domains E, D, A, B, or C can bind the Fc region of an antibody and can thereby be used as a ligand for purifying IgG antibodies. In addition, the combination of Linhult and Abrahmsén suggests making the G29A mutation in each of the domains because it would provide ligands that are less susceptible to protein degradation. Ex. 1004, 2; *see also* Ex. 1005, 2:33-37 ("[A] recombinant DNA fragment coding for any of the E D A B C domains of staphylococcal protein A, wherein the glycine codon(s) in the Asn-Gly coding constellation has been replaced by an alanine codon.").

Patent Owner contends that the G29A mutation would lead to a reduction in the Fab binding of domain C, and therefore, would lead away from making the mutation. PO Resp. 40–41 (citing Ex. 2010 at 2; Ex. 2011 at 25; Ex. 2012 at 25–26; Ex. 2013 at 2–3; Ex. 2025 ¶¶ 105–109; Ex. 2029 at 6–7). Patent Owner's cited references are directed to Fab binding. But claims 1 and 14 are not limited to Fab binding. Showing that the G29A mutation interferes with Fab binding says nothing about the ability of a mutated SPA domains E, D, A, B, or C to bind the Fc portion of IgG. *See, e.g.*, Ex. 2013, 3 ("The site responsible for Fab binding is structurally separate from the domain surface that mediates Fcɣ[[11]] binding.").

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[11] Fcɣ is the constant region of IgG involved in effector function. Specifically, "[t]he Fcɣ binding site has been localized to the elbow region at the CH2 and CH3 interface of most IgG subclasses, and this binding property has been extensively used for the labeling and purification of antibodies." Ex. 2013, 1. In other words, Fcɣ and Fc terminology are used interchangeably in the art.

Accordingly, we are not persuaded by Patent Owner's contention that the art teaches away from the combination as articulated by Petitioner.

(4) Additional Modifications

We are also not persuaded by Patent Owner's contention the ordinary artisan would not stop with a single G29A mutation in a SPA domain. *See* PO Resp. 43–45. Here, Abrahmsén expressly suggests making only a single mutation. Specifically, Abrahmsén contemplates "a recombinant DNA fragment coding for any of the E D A B C domains of staphylococcal protein A, wherein the glycine codon(s) in the Asn-Gly coding constellation has been replaced by an alanine codon" without additional mutations. Ex. 1005, 2:33–37.

(5) Summary

Having considered the evidence and argument cited by the Petition, which we adopt as our own, we are persuaded that Petitioner has shown by a preponderance of evidence of record that the combination of Linhult and Abrahmsén teach each of the limitations of claims 1 and 14. Petitioner not only has articulated a sufficient motivation for making the combination but has also established that there is a reasonable expectation of success for the binding of an IgG antibody to a SPA domain that contains an G29A mutation.

b) Claims 4 and 17

Petitioner argues that "[t]he 'capab[ility] of binding to the Fab part of an antibody,' as recited in claims 4 and 17, is an inherent property of the claimed C(G29A)-based SPA ligand." Pet. 28 (citing Ex. 1002 ¶¶ 120–127).

Petitioner contends that a person of ordinary skill in the art did not need to recognize the Fab binding property of Domain C to be motivated to select that domain for modification. "A POSA would not have been motivated only by Fab-binding ability,[] as even Dr. Bracewell agreed that 'a POSA would have understood that it was desirable to purify ***monoclonal antibodies*** for therapeutic use in 2006.'" Reply 8 (citing Ex. 1057, 75:17–76:4, 113:23–114:11, 157:24–158:9; Ex. 1061 ¶ 29).

Patent Owner argues that "[n]either Linhult itself nor the Petition provide any reason as to why the POSA would have 'plucked' Domain C from among the five listed SPA domain sequences (or in the figure, those plus Domain Z's sequence). PO Resp. 29 (citing *WBIP, LLC v. Kohler Co.*, 829 F.3d 1317, 1337 (Fed. Cir. 2016)); *see also* Sur-reply 1 ("Petitioners fail to show that the person of ordinary skill in the art (the 'POSA') would have plucked Domain C from the sea of prior art teaching a preference for Domains B and Z, substituted one and only one amino acid—the glycine at position 29 with an alanine (a 'G29A' mutation)—and used that mutated protein as a chromatography ligand for the purification of antibodies or antibody fragments.").

Patent Owner argues that

> [t]he very G29A amino acid substitution Petitioners now suggest the POSA would seek to employ with Domain C would have been known to have rendered Fab binding 'negligible' when implemented in Domain B. Ex. 2009 at 2; *see also, e.g.*, Ex. 2010 at 2 ("Fab binding activity is located to a region determined by helices 2-3, including the position mutated to yield the Z domain."); Ex. 2011 at 25 ("[I]t only takes a single residue change in SpA to eliminate either Fab or Fc binding. The sole difference in domain Z compared to domain B is the substitution of a glycine to an alanine . . . ."); Ex. 2012 at 25-26 ("[D]omain

> Z containing a single G29A-substitution compared to domain B exhibits little or no [Fab] binding. This might be due to the substitution since the C_β of the alanine would perturb the interaction between the two molecules.").

PO Resp. 40.

Claim 1 is directed to a composition. Claim 4 is dependent on claim 1 and further recites "wherein the ligand is capable of binding to the Fab part of an antibody." Ex. 1001, 15:56–57. The "capable of binding" language of claim 4, however, does not add any structural limitations to claim 1 it merely recites the function of the composition when used for example in an assay.[12]

"[T]erms [that] merely set forth the intended use for, or a property inherent in, an otherwise old [or obvious] composition . . . do not differentiate the claimed composition from those known in the prior art." *In re Pearson*, 494 F.2d 1399, 1403 (CCPA 1974). "Inherency may supply a missing claim limitation in an obviousness analysis. An inherent characteristic of a formulation [i.e. composition] can be part of the prior art in an obviousness analysis even if the inherent characteristic was unrecognized or unappreciated by a skilled artisan." *Persion Pharms. LLC v. Alvogen Malta Operations Ltd.*, 945 F.3d 1184, 1190 (Fed. Cir. 2019) (citations omitted).

As explained above (II.E.5.a), Abrahmsén teaches replacing the glycine codon in the Asn-Gly constellation in *any one* the SPA domains with an alanine codon. Ex. 1005, 2:21–23. Abrahmsén further explains that:

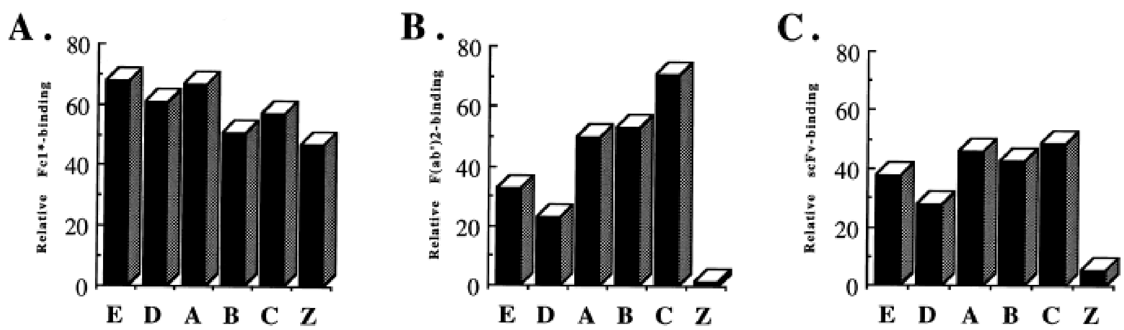
> The *Asn-Gly dipeptide* sequence is sensitive to hydroxylamine. As *this sequence is kept intact in all five IgG*

[12] The same issue applies to claims 14 and 17. We note that none of the claims in the '765 patent are directed to a method of using the composition in an assay to isolate Fab fragments.

> *binding domains of protein A* and as this amino acid sequence is present in the middle of an alpha helix involved in the binding to IgG (FIG. 2) there is very little chance to be successful in any amino acid change. . . . However, by simulating the Gly to Ala amino acid change in the computer we concluded that this change would not interfere with folding to protein A or binding to IgG.

Ex. 1005, 4:56–5:16 (emphasis added).

That the G29A mutation in domain B (resulting in domain Z) does not interfere with IgG binding was already established by Linhult. *See* Ex. 1004, 6 (Fig. 3 (showing IgG binding with domain Z). The Domain Z binding property to IgG is further supported by Jansson. Jansson Fig. 3, reproduced below, shows the side-by-side comparison of Fc1*[13], Fab, and scFv binding and confirms what was already suggested in Abrahmsén and Linhult – that a composition containing the G29A mutation in an SPA domain can bind IgG.



Panel A in Jansson Figure 3, reproduced above, shows that the single G29A mutation between Domain B and Domain Z does not result in the loss of IgG binding. Ex. 2029, 6 (Fig. 3). Because the other SPA domains are structurally so similar to Domain B there is a reasonable expectation that these domains would similarly retain the ability to bind IgG with the same G29A mutation. *See above* II.E.5.a.

[13] Fc1* is the constant region of human IgG1. Ex. 2029, 4. Fc1* is understood to be used as the IgG control in Jansson.

We, therefore, agree with Petitioner that the limitation of the Fab binding ability as recited in claim 4 is an inherent feature of the structure disclosed in the independent claim 1. *See* Pet. 28 (citing Ex. 1002 ¶¶ 120–123). For the reasons disclosed above (II.E.5.a), we find that Petitioner has shown by a preponderance of evidence that there is a reasonable expectation that making a G29A mutation in *any one* of the SPA domains would result in a structure that retains the ability to bind IgG.

(1) Unexpected Results

Patent Owner argues that objective indicia of non-obviousness of the Domain C–G29A based ligands requires reaching a conclusion of non-obviousness. PO Resp. 53–55. Specifically, Patent Owner contends that "the SPA ligands of the claimed chromatography matrices unexpectedly retained their ability to bind to the Fab part of an antibody despite the substitution of an alanine for the glycine at position 29 of the Domain C sequence." *Id.* at 55 (citing Ex. 2025 ¶ 123; Ex. 2030 at 18–19).

We are not persuaded by Patent Owner's unexpected results argument. The prior art does not need to recognize that Domain C retains the ability to bind Fab fragments after a G29A mutation. Petitioner's articulated rationale is that there is a reason to make the G29A mutations in *any one* of the SPA domains in order to get a product that is alkaline stable. Pet. 21–22 (citing Ex. 1002 ¶¶ 21, 99–11; Ex. 1004, 407–408, Fig. 1(a); Ex. 1006, 12; Ex. 1018 ¶ 29; Ex. 1019, 6:25–34). We find that Petitioner has shown by a preponderance of evidence that there is a reasonable expectation that making a G29A mutation in *any one* of the SPA domains results in a product that binds at least IgG. *See above* II.E.5.a. There is no requirement that the inherent characteristic of the Fab binding needed to be recognized in order to arrive at the conclusion that the recited structure in claim 1 would have been obvious. *See Persion Pharms.*, 945 F.3d at 1190 ("Our predecessor court similarly concluded that it 'is not the law' that 'a structure suggested by the prior art, and, hence, potentially in the possession of the public, is patentable ... because it also possesses an [i]nherent, but hitherto unknown, function which [the patentees] claim to have discovered.' *In re Wiseman*, 596 F.2d 1019, 1023 (C.C.P.A. 1979).").

c) Claims 3 and 16

Claim 3 depends from claim 1, and claim 16 depends from claim 14. Claims 3 and 16 recite the additional limitation "wherein the chromatography matrix has retained at least 95% of its original binding capacity after 5 hours incubation in 0.5 M NaOH." Ex. 1001, 15:52–56. We find Petitioner has shown by a preponderance of the evidence that the combination of Linhult, Abrahmsén, and Hober teaches the additional limitations of the dependent claims for the reasons stated in the Petition which we adopt as our own. *See* Pet. 25–28, 30; Reply 20–21.

We are not persuaded by Patent Owner's argument that "none of Petitioners' cited references actually describe a C(G29A)-based SPA ligand, let alone provide alkaline stability data or test results concerning the same, the POSA is simply left to guess at how such a ligand would perform." PO Resp. 49 (citing Ex. 2025 ¶¶ 121–124).

Petitioner has shown by a preponderance of the evidence of record that there is a reason for making the G29A mutation in *any one* of the four remaining SPA domains in order to produce a SPA product that is more alkaline stable and would reasonably bind IgG. We, therefore, agree with Petitioner, that the ability to retain the requisite binding capacity after a period of exposure to alkaline treatment is a property of the composition. Pet. 25–28; *see* Ex. 1002, ¶¶115–119; Ex. 1004, 6 ("Figure 3, the Z(N23T) mutant shows higher resistance to alkaline conditions than the Z domain when exposed to high pH values."). That Linhult recognizes that additional mutations could further improve alkaline stability does not detract from Linhult's teaching that a composition containing the single G29A mutation in SPA Domain B retains IgG binding. Ex. 1004, 6, *see id.* at 4 ("The Z-domain already possesses a significant tolerance to alkaline conditions.").

d) Claims 2, 12, 15, and 25

Claims 2 and 12 depend from claim 1, and claims 15 and 25 depend from claim 14. We find Petitioner has shown by a preponderance of the evidence that the combination of Linhult, Abrahmsén, and Hober teaches the additional limitations of the dependent claims for the reasons stated in the Petition which we adopt as our own. *See* Pet. 25–28, 30.

We have reviewed Petitioner's arguments and the underlying evidence cited in support and determine that Petitioner establishes that of Linhult, Abrahmsén, and Hober teaches the additional limitations of these dependent claims. Specifically, Petitioner asserts that Linhult describes multimerization of the ligand as recited in claims 2, 12, 15; and 25. Pet. 26 (citing Ex. 1004, 410; Ex. 1002 ¶¶112–114), *see id.* at 30 (citing Ex. 1002 ¶¶ 128–129).

Patent Owner does not offer arguments addressing Petitioner's substantive showing with respect to claims 2, 12, 15, and 25. *See generally* PO Resp.

6. Summary

For the foregoing reasons, we determine that Petitioner has shown by a preponderance of evidence that of claims 1–7, 10–20, and 23–26 of the ’765 patent are unpatentable based on the combination of Linhult, Abrahmsén, and Hober.

F. *Other Asserted Grounds*

Petitioner also asserts that claims 1–4, 12, 14–17, and 25 are unpatentable as obvious over Linhult and Abrahmsén (Pet. 18–30); claims 1–7, 10–20, and 23–26 are unpatentable as obvious over Linhult and Hober (*id.* at 30–48); claims 1–7, 10–20, 23–26 are unpatentable as obvious over Abrahmsén and Hober (*id.* at 49–60); claims 1–7, 10–20, and 23–26 are unpatentable over Berg and Linhult (’043 IPR Pet. 20–33); claims 2, 3, 15, and 16 are unpatentable over Berg, Linhult, and Hober (*id.* at 33–38); claims 1, 2, 5–7, 10–15, 18–20, and 23–26 are unpatentable over Berg and Abrahmsén (*id.* at 38–44); and claims 2–4, 15–17 are unpatentable over Berg, Abrahmsén, and Hober (*id.* at 44–46) under 35 U.S.C. §103(a). However, because Petitioner has already shown that the challenged claims 1–7, 10–20, and 23–26 are unpatentable over Linhult, Abrahmsén, and Hober as obvious, as discussed supra, we do not reach these additional asserted grounds. *See Beloit Corp. v. Valmet Oy*, 742 F.2d 1421, 1423 (Fed. Cir. 1984) (“The Commission . . . is at perfect liberty to reach a ‘no violation’ determination on a single dispositive issue.”); *Boston Sci. Scimed, Inc. v. Cook Grp., Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020) (recognizing that ‘[t]he Board has the discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims”).

III. CONCLUSION[14]

For the foregoing reasons, we determine that Petitioner has demonstrated by a preponderance of the evidence that claims 1–7, 10–20, and 23–26 of the ’765 patent are unpatentable on the bases set forth in the following table.

In summary:

Claim(s)	**35 U.S.C. §**	**Reference(s)/Basis**	**Claim(s) Shown Unpatentable**	**Claim(s) Not shown Unpatentable**
1–4, 12, 14–17, 25	103(a)	Linhult, Abrahmsén[15]		
1–7, 10–20, 23–26	103(a)	Linhult, Hober[16]		
1–7, 10–20, 23–26	103(a)	Linhult, Abrahmsén, Hober	1–7, 10–20, 23–26	
1–7, 10–20, 23–26	103(a)	Abrahmsén, Hober[17]		

[14] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

[15] As explained above, we do not reach this ’036 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of all the challenged claims.

[16] *See supra*, n.15.

[17] *See supra*, n.15.

Claim(s)	**35 U.S.C. §**	**Reference(s)/Basis**	**Claim(s) Shown Unpatentable**	**Claim(s) Not shown Unpatentable**
1-7, 10–20, 23–26	103(a)	Berg, Linhult[18]		
2, 3, 15, 16	103(a)	Berg, Linhult, Hober[19]		
1, 2, 5–7, 10–15, 18–20, 23–26	103(a)	Berg, Abrahmsén[20]		
2–4, 15–17	103(a)	Berg, Abrahmsén, Hober[21]		
Overall Outcome			1–7, 10–20, 23–26	

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that the preponderance of the evidence of record has shown that claims 1–7, 10–20, and 23–26 of the ’765 patent are found unpatentable; and

FURTHER ORDERED because this is a final written decision, the parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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[18] As explained above, we do not reach this ’043 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of all the challenged claims.
[19] *See supra*, n.18.
[20] *See supra*, n.18.
[21] *See supra*, n.18.

FOR PETITIONER:

Naveen Modi
Eric Dittmann
Phillip Citroen
Isaac Ashkenazi
PAUL HASTINGS LLP
naveenmodi@paulhastings.com
ericdittmann@paulhastings.com
phillipcitroen@paulhastings.com
isaacashkenazi@paulhastings.com

FOR PATENT OWNER:

David Krinsky
David Berl
WILLIAMS & CONNOLLY LLP
dkrinsky@wc.com
dberl@wc.com